UNIVERSITAS MERCU BUANA

Available online at: http://publikasi.mercubuana.ac.id/index.php/ijiem

IJIEM (Indonesian Journal of Industrial Engineering & Management)

ISSN (Print) : 2614-7327 ISSN (Online) : 2745-9063

IJIEM

Application of Economic Order Quantity (EOQ) Method to Improve the Cost Efficiency of Laundry consumables Inventory (Study Case at Kumala Siwi Mijen General Hospital)

Rubiyayanti*

Logistics Management Department, Rukun Abdi Luhur Polytechnic, Jl. Raya Kudus-Jepara No. Km. 6, Madaran, Mijen, Kaliwungu, Kudus, 59316, Indonesia

ARTICLE INFORMATION

Article history:

Received:
Revised:
Accepted:

Category: select one (Research paper, Review paper, View point, Technical paper, Conceptual paper)

Keywords:
Economic Order Quantity
Cost efficiency
Laundry consumables

*Corresponding Author
Xxxxx (Name)
E-mail: author@affiliation.xx.xx

A B S T R A C T

Kumala Siwi Mijen General Hospital often faces the problem of laundry consumables inventory, both shortages and excesses. This study uses the Economic Order Quantity (EOQ) method to plan inventory and reduce ordering costs. The analysis method used is descriptive with quantitative data .

The results show that the application of EOQ can save ordering costs up to 32%. For example, for laundry consumables Brodklin Liquid, the ordering cost is reduced by Rp. 234,084, for Brodklin Manual it is reduced by Rp. 150,483, for Brodklin Matik it is reduced by Rp. 171,982, and for Brodklin Fragrance it is reduced by Rp. 229,308. Total cost savings from Rp. 2,469,839 to Rp. 1,683,582 after using EOQ.

1. INTRODUCTION

The rapid economy and advances in science and technology have led to increasingly fierce competition between hospitals. Effective management is key in making decisions and controlling hospital activities (Parahita et al., 2018). Kumala Siwi Mijen General Hospital requires good ordering of raw materials to maintain a smooth production process. Inventories, including raw materials, semi-finished materials, and finished goods, are important to meet customer needs and satisfaction, as well as anticipate urgent needs.

The use of laundry consumables at Kumala Siwi Mijen General Hospital (RSU Kumala Siwi Mijen) varies monthly depending on the number of patients, often resulting in inaccuracies in inventory planning. Inventory shortages hamper distribution, while overstocks increase storage and operational costs.

To solve this problem, the Economic Order Quantity (EOQ) method can be applied. EOQ helps determine the most economical purchase quantity, reduce costs and maintain optimal inventory levels (Prawirosentono, 2021). In addition, setting the right reorder point (ROP) is important to prevent shortages and accumulation of raw materials (Wasis, 1997). This study entitled "Application of the Economic Order Quantity (EOQ) Method to Increase the Efficiency of laundry consumables Inventory Costs: Case Study at RSU Kumala Siwi Mijen" aims to evaluate how EOQ can improve the efficiency of inventory management in hospitals.

However, many hospitals, including RSKS, face problems with inventory, such as excess or shortage of raw materials, which can hinder the production process and increase storage costs. RSKS, which manages laundry raw materials for washing hospital linen, experiences irregularity in ordering raw materials due to fluctuations in the number of patients. This leads to non-optimal inventory and disrupts the smooth production process.

2. LITERATURE REVIEW

2.1 Inventory

Inventory is a crucial aspect of hospital management, covering all activities related to ordering, storing, using, and inventorying goods. Alexandri (2009) defines inventory as an asset that includes goods for sale, goods in the production process, and raw materials. Dwiky Guntara et al. (2020) add that inventory involves stocks of goods managed by the organization to meet demand from both internal and external customers. Different types of organizations have different types of inventory, such as department stores that store merchandise, plant stock at nurseries, or football clubs that manage player inventory.

2.2 Factors Affecting Inventory

According to Prawirosentono (2001), there are several main factors that affect inventory management, namely:

- Estimated Usage: Estimated raw material requirements based on the planned production period.
- Raw Material Prices: The price of raw materials plays an important role in determining the amount of inventory that needs to be prepared.
- Inventory Costs: Includes ordering costs and storage costs that must be taken into account in inventory management.
- Lead Time: The duration from ordering to receiving goods in the warehouse affects inventory management.

Sugiyono (2016) emphasizes the importance of inventory control to ensure efficient ordering from the best sources to maintain inventory integrity and prevent damage or loss of goods.

2.3 Inventory Cost

Freddy Rangkuty (2011) divides inventory costs into several main categories:

- Storage Costs: Costs associated with storing goods, including maintenance and warehouse rent.
- Ordering Costs: Costs incurred each time an order is placed, covering the administration and management of the order.
- Setup Costs: Costs incurred when goods are produced internally.
- Shortage Cost: Costs incurred due to insufficient inventory to meet demand.

Saragi and Setyorini (2014) classify the costs in the inventory system as:

- Purchase cost: Price per unit of goods.
- Procurement Costs: Costs associated with ordering and manufacturing goods.
- Storage Costs: Costs associated with storing goods in a warehouse.
- Inventory Shortage Cost: Costs that arise when inventory cannot meet demand.

2.4 Economic Order Quantity (EOQ)

The Economic Order Quantity (EOQ) method is an inventory control technique designed to minimize total ordering and storage costs. Handoko (2014) explains that EOQ, or the fixed-order-quantity model, aims to determine the optimal order quantity to reduce total costs. EOQ keeps the amount of inventory ordered consistent each time it reaches the reorder point.

a. EOQ Policy

According to Herjanto (2008), the policies or assumptions underlying the use of the EOQ model include:

- The item ordered is one of a kind.
- Demand for goods is constant and known.
- Ordering costs and storage costs are fixed and known.
- Goods are received in one batch.
- The item price is fixed and does not depend on the quantity purchased.
- The lead time is known and fixed.

b. EOQ Determination

The determination of EOQ involves the calculation of several types of costs:

- Order Costs: Costs associated with ordering activities, including preparation, shipping, and invoicing costs. The formula for calculating order cost is:

$$\text{Order Cost} = \frac{D}{Q} \text{ x } S$$

Where Q is the number of items per order, D is the annual demand, and S is the ordering cost per time.

Storage Costs: Costs associated with storing goods, including maintenance, insurance, and building rent. The formula for calculating storage costs is:

$$\text{Storage Cost} = \frac{Q}{2} \text{ x } H$$

Where H is the storage cost per unit per year.

3. RESEARCH METHOD

This study uses a quantitative descriptive approach to analyze the inventory phenomenon at RSU Kumala Siwi Mijen. The research objective is to describe current conditions through in-depth and comprehensive analysis, and process data quantitatively with a mathematical/statistical approach (Nazir, 2009).

Data sources used Interviews were conducted to obtain primary data by means of direct question and answer between researchers and the head of household affairs in the logistics unit of Kumala Siwi Mijen Hospital and Documentation involved collecting secondary data from company records regarding the number of purchases of laundry laundry consumables and data related to inventory management.

3.1 Economic Order Quantity

The EOQ formula is used to determine the optimal order quantity:

$$IQ = \sqrt{(2SD/H)} / \sqrt{(2SD/I \text{ x } C)}$$

Description:

Q = Economical quantity per order
D = The number needs in units (unit) per year
S = Ordering Cost for Once Order
H = Storage cost per unit per year
I = % storage cost
C = Purchase price

3.2 Total Inventory Cost (TC)

The total inventory cost is calculated by the formula:

$$\text{Total Cost (TC)} = \frac{D}{Q} \text{ x } S + \frac{Q}{2} \text{ x } H$$

Description:

TC = total cost of inventory.
Q = quantity of goods per order.
D = annual demand for inventory items in units per year.
S = order cost for each order.
H = storage cost per unit per year.

3.3 Safety Stock (SS)

Safety stock is calculated to reduce the risk of running out of goods:

SS = Lead Time x Average Daily Usage

3.4 Reorder Point (ROP)

ROP is calculated to determine when to reorder, using the formula:

ROP = Lead Time x Average Consumption

4. RESULT AND DISCUSSION

The results of this study include an analysis of a logistics warehouse that contains various types of laundry consumables, which include liquid brodklin, manual brodklin, matik brodklin, fragrance brodklin. Orders are placed once a month. In 2023 Kumala Siwi Mijen Kudus Hospital placed an order for laundry consumables, which includes brodklin liquid, brodklin manual, brodklin matik, brodklin fragrance. The order quantity and average inventory level based on what is needed in 2023 is shown in the following table:

Table 1. Laundry Consumables Ordering Year 2023

Material Name	**Order/year (liters)**
Brodklin Liquid	1.225
Brodklin Manual	1.050
Brodklin Matik	1.800
Brodklin Fragrance	1.200
Total	**5.275**

Source: RSU Kumala Siwi Mijen Kudus

Based on the laundry consumables Laundry order plan, the logistics department can estimate the number of laundry consumables Laundry needs used.

Table 2. Laundry consumables Usage in 2023

Material Name	**Usage/year (Liters)**
Brodklin Liquid	1.150
Brodklin Manual	1.070
Brodklin Matik	1.685
Brodklin Fragrance	1.170
Total	**5.075**

Source: RSU Kumala Siwi Mijen

4.1 Inventory Cost

Inventory cost includes the total expenditure on buying and storing laundry consumables. This includes the cost of ordering and storing goods. The larger the inventory stored, the higher the storage cost. Similarly, the more frequently orders are placed, the higher the ordering cost. The following is a breakdown of the total ordering cost of RSKS laundry consumables in 2023. Ordering costs include telephone and administration costs, while transportation costs are included in the price of the goods and correspondence costs do not exist because orders are placed over the phone.

Table 3. Inventory Cost

Component	**Cost Type**	**Amount/Year (RP)**
Ordering Cost	Telephone charges	12.000
	Administration fee	240.000
	Total cost	**252.000**
Storage Cost	Electricity Cost Log gdg area 10.000.000/420= 23.810 laundry consumables Laundry shelf area 3,5 x 23.810 = 83.335	1.000.020
	Shelf depreciation cost 2.500.000-25%	625.000
	Salary cost 3,5/80 x 24.000.000	1.050.000
	Total cost	**2.675.020**

Source: RSU Kumala Siwi Mijen

$$\text{Storage cost per liter} = \frac{\text{Total storage cost}}{\text{Total Purchase}}$$

$$= \frac{2.675.020}{5.275}$$

$$= 507$$

Liquid storage costs

$$\% \text{ Cost} = \frac{\text{Storage Cost}}{\text{Total Storage Cost}} \text{ x } 100\%$$

$$= \frac{507 \text{ x } 1.150}{2.675.020} \text{ x } 100\%$$

$$= 22\%$$

Manual storage cost

$$\% \text{ Cost} = \frac{\text{Storage Cost}}{\text{Total Storage Cost}} \text{ x } 100\%$$

$$= \frac{507 \text{ x } 1.070}{2.675.020}$$

$$= 20\%$$

Storage cost of an electric

$$\% \text{ Cost} = \frac{\text{Storage Cost}}{\text{Total Storage Cost}} \text{ x } 100\%$$

$$= \frac{507 \text{ x } 1.685}{2.675.020}$$
$$= 32\%$$

Fragrance storage cost

$$\% \text{ Cost} = \frac{\text{Storage Cost}}{\text{Total Storage Cost}} \text{ x } 100\%$$
$$= \frac{507 \text{ x } 1.170}{2.675.020} \text{ x } 100\%$$
$$= 22\%$$

By applying the EOQ Method, hospitals can determine the economical ordering quantity of raw materials, based on constant demand and lead time.

Laundry consumables for Brodklin Liquid is :

$$\text{EOQ} = \sqrt{\frac{2SD}{I \text{ x } C}}$$
$$= \sqrt{\frac{2 \text{ x } 58.521 \text{ x } 1.150}{22\% \text{ x } 16.150}}$$
$$= \sqrt{\frac{134.598.300}{3.553}}$$
$$= \sqrt{37.883} = 195 \text{ liters}$$

$$\text{Order frequency} = \frac{1.150}{195}$$
$$= 5,8 \longrightarrow 6 \text{ times}$$

Based on the EOQ calculation for laundry consumables for Brodklin Liquid, it is found that the optimal order quantity in 2023 is 195 liters with an order frequency of 6 times.

Laundry consumables Brodklin Manual is :

$$\text{EOQ} = \sqrt{\frac{2SD}{I \text{ x } C}}$$
$$= \sqrt{\frac{2 \text{ x } 50.161 \text{ x } 1.070}{22\% \text{ x } 19.000}}$$
$$= \sqrt{\frac{107.344.540}{3.800}}$$
$$= \sqrt{28.249} = 168 \text{ liters}$$

$$\text{Order frequency} = \frac{1.070}{168}$$
$$= 6,4 \longrightarrow 6 \text{ times}$$

Based on the EOQ calculation for laundry consumables Laundry Brodklin Manual, 168 liters with an order frequency of 6 times.

Laundry consumables for Brodklin Matic is :

$$\text{EOQ} = \sqrt{\frac{2SD}{I \text{ x } C}}$$
$$= \sqrt{\frac{2 \text{ x } 85.991 \text{ x } 1.686}{22\% \text{ x } 19.000}}$$
$$= \sqrt{\frac{289.789.670}{6.080}}$$
$$= \sqrt{47.663} = 218 \text{ liters}$$

$$\text{Order frequency} = \frac{1.685}{218}$$
$$= 7,7 \longrightarrow 8 \text{ times}$$

Based on the EOQ calculation for laundry consumables Laundry Brodklin Matik, there are 218 liters with an order frequency of 8 times.

Laundry consumables for Brodklin Fragrance is :

$$\text{EOQ} = \sqrt{\frac{2SD}{I \text{ x } C}}$$
$$= \sqrt{\frac{2 \text{ x } 57.327 \text{ x } 1.170}{22\% \text{ x } 18.050}}$$
$$= \sqrt{\frac{134.145.180}{3.971}}$$
$$= \sqrt{33.781} = 184 \text{ liters}$$

$$\text{Order frequency} = \frac{1.170}{184}$$
$$= 6,4 \longrightarrow 6 \text{ times}$$

Based on the EOQ calculation for laundry consumables Laundry Brodklin deodorizer, 184 liters with an order frequency of 6 times.

4.2 Reorder Point

ROP can be calculated using the following formula:
ROP = Sefety Stock (SS) + (Lead time x Daily quantity)
SS = lead time x average daily usage

ROP for Brodklin Liquid

SS = lead time x average
= 7 X (1.150/365) = 22
ROP = SS + (Lead time x Daily quantity)
= 22 + 22 = 44

ROP for Manual Brodklin
SS = lead time x average
= 7 X (1.070/365) = 21
ROP = SS + (Lead time x Daily quantity)
= 21 + 21 = 42

ROP for Brodklin matic
SS = lead time x average
= 7 X (1.685/365) = 32
ROP = SS + (Lead time x Quantity days)
= 32 + 32 = 64

ROP for Brodklin Fragrance
SS = lead time x average
= 7 X (1.170/365) = 22
ROP = SS + (Lead time x Daily quantity)
= 22 + 22 = 44

From the data above, the safety stock for Brodklin Liquid is 22 liters, Brodklin Manual 21 liters, Brodklin Matic 32 liters, and Brodklin Fragrance 22 liters. The ROP (Reorder Point) is 44 liters each for Brodklin Liquid and Fragrance, 42 liters for Brodklin Manual, and 64 liters for Brodklin Matic.

Table 4. Conclusion of All Calculation

Item Name	Order Cost	EOQ	Frequency Before	Frequency After	ROP	Before EOQ	After EOQ	Difference
Brodklin liquid	58521	195	10	6	44	585.210	351.126	234.084
Brodklin Manual	50161	168	9	6	42	451.449	300.966	150.483
Brodklin Matik	85991	218	10	8	64	859.910	687.928	171.982
Brodklin Fragrance	57327	184	10	6	44	573.270	343.962	229.308
Total	**252.000**	**765**	**39**	**26**	**194**	**2.469.839**	**1.683.982**	**785.857**

The data shows that before using the EOQ method, the total ordering cost was Rp 2.469.839, while after applying EOQ, the cost dropped to Rp 1.683.982, resulting in savings of Rp 785.857.
This difference is due to the previous higher ordering frequency 39 times per year-compared to 26 times per year with the EOQ method. The hospital previously placed orders more frequently to reduce the risk of stock-outs and maintain a smooth production process.

5. CONCLUSION

The current inventory management of laundry consumables at RSU Kumala Siwi Mijen uses a monthly order frequency with a total order of 5.275 liters in 2023.
The Economic Order Quantity (EOQ) method can significantly reduce order frequency and total inventory costs.
EOQ provides an optimal ordering quantity with a lower frequency than the current method.
The application of EOQ can save inventory costs of IDR 785.857 per year (32%), with costs before EOQ IDR 2.469.839 and after EOQ IDR 1.683.982.

REFERENCES

Alexandri, B. (2009). Manajemen Keuangan Bisnis. Edisi kedua. Bandung. Alfabeta.
Dwiky Guntara, Muhammad Irwan Padli Nasution, dan Adnan Buyung Nasution. 2020. Implementasi

Metode Economic Order Quantity Pada Aplikasi Pengendalian Bahan Produksi Sandal Mirado . Jurnal Teknik Informatika Vol. 13 No. 1

Guntara, D., Nasution, M. I. P., & Nasution, A. B. (2020). Implementasi metode *economic order quantity* pada aplikasi pengendalian bahan produksi sandal Mirado. *Jurnal Teknik Informatika*, *13*(1), 31-42.

Handoko, H T. 2014. Dasar-dasar manajemen produksi dan operasi. Edisi pertama. Yogyakarta: BPFE – Yogyakarta.

Heizer, Jay dan Barry Render. 2010. Manajemen operasi. Edisi Sembilan buku 2.Jakarta: Salemba empat.

Herjanto, Hery. 2008.Manajemen operasi. Edisi ketiga. Cetakan ketujuh. Jakarta:Grasindo.

Nazir, Moh. 2009. Metode Penelitian. Bogor : Ghalia Indonesia.

Palupi, P. M., Korawijayanti, L., & Handoyono, R. (2018). Penerapan Metode *Economic Order Quantity (EOQ)* Untuk Meningkatkan Efisiensi Biaya Persediaan Bahan Baku (Studi Kasus pada PT Nusamulti Centralestari). In *Prosiding Seminar Nasional Unimus* (Vol. 1).

Prawisentono, Suryadi. 2001. Manajemen operasi: analisis dans tudikasus. Edisi ketiga. Jakarta: BumiAksara.

Rangkuti, Freddy. 2011. Manajemen persediaan aplikasi di bidang bisnis. Cetakan keenam. Jakarta : Raja Grafindo Persada.

Rangkuti, Freddy. 2011. Manajemen persediaan aplikasi di bidang bisnis.Cetakan keenam. Jakarta : Raja Grafindo Persada.

Meningkatkan Efisiensi Pengendalian Persediaan Bahan Baku Pada CV Syahdika. *Perspektif: Jurnal Ekonomi dan Manajemen Akademi Bina Sarana Informatika*, *19*(2), 158-164.

Saragi, Lestari Gema. 2014. Analisis Pengendalian Persediaan Bahan Baku Daging Dan Ayam Dengan Menggunakan Metode *Economic Order Quantity (EOQ)* Pada Restoran Steak Ranjang Bandung. Jurnal E-Proceding ISSN 2355-9357 Vol.1 No. 3 Hal 542-553.

Serius Laoli,Kurniawan Sarototonafo Zai dan Natalia Kristiani Lase (2022). Penerapan Metode Economic Order Quantity (Eoq), Reorder Point (Rop), Dan Safety Stock (Ss) Dalam Mengelola Persediaan Di Grand Katika Gunungsitoli. Jurnal EMBA Vol. 10 No. 4 Oktober 2022, Hal. 1269-1273

Slamet, (2007). Kiat Meningkatkan Kinerja, Cetakan Pertama,. ; Jakarta: Rineka Cipta.

Sugiyono. 2016. Metode Penelitian Pendidikan. Bandung : CV Alfabeta.

Suryani dan Dedi Can (2018). Analisis Pengendalian Persediaan Sistem Manajemen Persediaan Bahan B0aku Kemasan. Jurnal INTECH Teknik Industri Universitas Serang Raya Vol 4

Tinangon Joshua Timothy dan Jacky Sumarauw (2020). Analisis Pengendalian Persediaan Bahan Baku Kemasan Plastik Pada Pt. Asegar Murni Jaya Desa Tumaluntung Kab. Minahasa Utara. Jurnal EMBA Vol.8 No.1 Januari 2020, Hal. 2180 – 2188

Usuli, Sudarto. 2013. Analisis *Economic Order Quantity (EOQ)* pada Perusahaan Tahu Tempe Vira. Jurnal EKOMEN vol. 13 no. 2

Yuli Evitha dan Fauzy Ma'ruf HS (2019). Pengaruh Penerapan Metode Economic Order Quantity (EOQ) Terhadap Pengendalian Persediaan Bahan Baku Produksi di PT. Omron Manufacturing Of Indonesia. Jurnal Logistik Indonesia Vol. 3, No. 2

Wasis (1997). Pengantar Ekonomi Perusahaan. Bandung: Alumni